

PROFILE

Fourth-year chemical engineering student interested in chemical reactions and sustainable engineering. Experience across team-based plant design, materials and energy projects, laboratory work, scientific programming, computational simulation, and educational outreach. Particularly interested in materials science, renewable energy, and green chemistry.

CURRENT EXPERIENCE

Intern

CG Brewery · Nawalparasi, Nepal

SELECTED PROJECTS

CO₂ Utilization for Green Methanol

Fourth-Year Plant Design · KU

Contributed to a team conceptual plant-design study and Aspen Plus simulation integrating cement-flue-gas CO₂ capture, alkaline electrolysis, reverse water-gas shift, methanol synthesis, recycle, and two-stage distillation. Scope included mass and energy balances, equipment and instrumentation, process diagrams, and techno-economic evaluation.

MSWI Fly Ash Utilization in Cement Clinker Production

2025

Assessed the feasibility, techno-economics, environmental impact, and circular-economy potential of using industrial fly ash as a clinker substitute.

Geothermal Energy — Air-Based Ground Source Heating

2024 · WEIR Lab

Designed a renewable geothermal heating system; analysed heat transfer and optimisation, and evaluated thermal performance and economic viability.

Photocatalytic Decomposition of Ethylene

2024

Investigated catalytic ethylene removal for post-harvest shelf life using controlled experiments and catalyst-performance optimisation.

Biodegradable Plastic Film from Potato Starch

2023

Synthesised bio-based polymer films and assessed mechanical, thermal, and biodegradation properties against conventional plastics.

EDUCATION

B.E. Chemical Engineering Kathmandu University · 4th year · Expected 2026	2022—present
School Leaving Certificate (+2) KU High School	2022
Secondary Education Examination Siddhartha School and College	2019

LEADERSHIP & COMMUNITY

Electronics Camp Organiser Independent initiative Delivered a three-month embedded systems, sensors, and actuators curriculum; mentored school students on C/C++ projects including automated lighting and obstacle-avoiding robots.	2024
Member · KUCONC Kathmandu University Circle of Noble Chemineers	2022—present
Assistant Teacher · Mathematics & Science Siddhartha Secondary School	2017—2020

AWARDS & CERTIFICATES

Japanese Language Scholarship FORDAYS Co. Ltd., Japan · USD \$2,000 award and industry visit	2024
Python for Web App Development Course certification	2020
Data Management for Cyber Security Course certification	2020

TECHNICAL SKILLS

Programming & data Python, NumPy, Pandas, Matplotlib, Scikit-learn, C, C++
Process & simulation Aspen Plus, plant design, P&ID, reactor design
CAD & laboratory AutoCAD, SolidWorks, Fusion 360, materials characterisation

LANGUAGES

English · Fluent Nepali · Native Hindi · Conversational Japanese · JLPT N5